Large Language Models and How ChatGPT-style Chatbots Work

Knowledge document 004 of 100

Introduction to Large Language Models

A Large Language Model (LLM) is an artificial intelligence model trained on enormous amounts of text so that it can understand and generate human language. ChatGPT from OpenAI, Claude from Anthropic, Gemini from Google, Llama from Meta and DeepSeek are well-known examples. These models can answer questions, summarise documents, translate languages, write essays and computer code, and hold conversations. They are called "large" because they contain billions or even trillions of parameters, the numerical values learned during training, and because they are trained on trillions of words from books, websites, articles and code.

Brief History

Earlier language systems used hand-written rules or simple statistics. The breakthrough came in 2017 when Google researchers published the paper "Attention Is All You Need", introducing the Transformer architecture. In 2018 OpenAI released the first GPT (Generative Pre-trained Transformer) and Google released BERT. GPT-2 followed in 2019 and GPT-3, with 175 billion parameters, in 2020. On 30 November 2022 OpenAI launched ChatGPT, which reached an estimated 100 million users within about two months and began the global generative AI boom. Since then, many companies have released competing models, and models have become multimodal, meaning they can work with images, audio and video as well as text.

What Is a Token?

A token is the basic unit of text that a language model reads and writes. A token is not exactly a word: it can be a whole word, part of a word, a punctuation mark or a space. For example, a short common word like "the" is usually one token, while a long or rare word like "unbelievable" may be split into several tokens. In English, one token is roughly four characters, or about three-quarters of a word, so 100 tokens is about 75 words. Before processing, a tokenizer converts text into tokens and then into numerical IDs. Models have a context window, which is the maximum number of tokens they can consider at once, including both the prompt and the response. Cost and speed of using LLMs through an API are normally measured in tokens.

What Is a Prompt?

A prompt is the input text or instruction that a user gives to a language model to get a response. A prompt can be a question, a command, an example or a long document with instructions. The quality of the answer depends heavily on the quality of the prompt, and the skill of writing effective prompts is called prompt engineering. Good prompts are clear, specific and give context, a role, the desired format and examples. Techniques include zero-shot prompting (no examples), few-shot prompting (a few examples included), and chain-of-thought prompting (asking the model to reason step by step). A system prompt is a hidden instruction set by the developer that defines the chatbot's behaviour, tone and rules.

How ChatGPT-style Chatbots Work

At its core, an LLM is a next-token predictor. Given the tokens so far, it calculates a probability for every possible next token and picks one, then repeats the process again and again to build a full answer. The Transformer uses a mechanism called self-attention, which lets the model weigh how relevant every earlier token is to the token being produced, so it can capture context and long-range meaning. Building a chatbot involves several stages. In pre-training, the model reads massive text collections and learns grammar, facts and reasoning patterns by predicting missing or next tokens. In fine-tuning (instruction tuning), it is trained on example conversations so that it follows instructions. In alignment, often using Reinforcement Learning from Human Feedback (RLHF), human reviewers rank answers and the model learns to prefer helpful, honest and safe responses. A setting called temperature controls randomness: low temperature gives focused, predictable answers while high temperature gives more creative, varied ones.

Why Do LLMs Sometimes Hallucinate?

A hallucination is when a language model produces text that sounds confident and fluent but is false, invented or unsupported, such as a fake book title, a wrong date or a non-existent legal case. LLMs hallucinate because they are designed to predict the most plausible next words, not to check facts against a database of truth. They do not truly know what is true; they learn statistical patterns in language. Other causes include gaps or errors in the training data, knowledge that stops at a training cut-off date, ambiguous or misleading prompts, and the model's tendency to give an answer rather than admit uncertainty. Hallucinations can be reduced by Retrieval-Augmented Generation (RAG), which gives the model trusted documents to answer from, by asking for sources, by lowering the temperature, by clear prompts and by human review of important outputs.

Uses and Indian Examples

LLMs are used for customer-support chatbots, coding assistants, document summarisation, legal and medical research support, education and tutoring, content writing and data analysis. In India, models and apps are being adapted for Indian languages; Bhashini and several startups provide translation and voice tools for Tamil, Hindi, Telugu and other languages, and Indian companies are building AI chatbots for banking, railways booking and government services. Colleges in Chennai and across India use LLMs for teaching and research support.

Challenges and Future

Challenges include hallucinations, bias inherited from training data, privacy and copyright concerns, high energy and computing costs, and misuse such as deepfakes or cheating. The future includes smaller efficient models that run on laptops and phones, better reasoning, AI agents that use tools and complete tasks, stronger support for regional languages and wider use of RAG with private data.

Quick Facts

- An LLM is a transformer-based model trained on huge amounts of text to understand and generate language.
- A token is a small unit of text, roughly four characters or three-quarters of an English word.
- A prompt is the input instruction or question given to a model.
- ChatGPT was launched by OpenAI on 30 November 2022.
- Hallucination happens because LLMs predict plausible words rather than verifying facts.